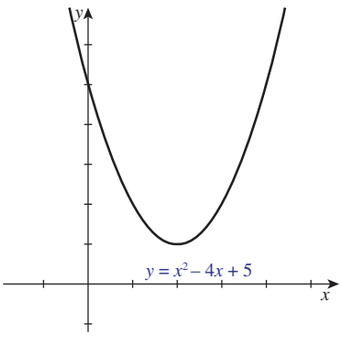


Figure 2.4

Discriminant	Nature of roots	Parabola
Positive	real and distinct	intersects x-axis at two points
Zero	real and equal	touches x-axis at one point
Negative	no real roots	does not meet x-axis

Example 2.10 If a and b are the roots of the equation $x^2 - px + q = 0$, find the value of $\frac{1}{a} + \frac{1}{b}$.

Solution:
Given that a and b are the roots of $x^2 - px + q = 0$. Then, $a + b = p$ and $ab = q$. Thus,
$\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} = \frac{p}{q}$.

Example 2.11 Find the complete set of values of a for which the quadratic $x^2 - ax + a + 2 = 0$ has equal roots.

Solution:
The quadratic equation $x^2 - ax + a + 2 = 0$ has equal roots.
So, its discriminant is zero. Thus, $D = b^2 - 4ac = a^2 - 4a - 8 = 0$.
So, $a = \frac{4 \pm \sqrt{48}}{2}$ which gives $a = 2 + \sqrt{12}, \quad 2 - \sqrt{12}$.

Example 2.12 Find the number of solutions of $x^2 + |x - 1| = 1$.

Solution:

Case (1). For $x \geq 1$, $|x - 1| = x - 1$.
Then the given equation reduces to $x^2 + x - 2 = 0$. Factoring we get $(x+2)(x-1) = 0$, which implies $x = -2$ or 1. As $x \geq 1$, we obtain $x = 1$.

Case (2). For $x < 1$, $|x - 1| = 1 - x$
Then the given equation becomes $x^2 + 1 - x = 1$. Thus we have $x(x - 1) = 0$ which implies $x = 0$ or $x = 1$. As $x < 1$, we have to choose $x = 0$.

Thus, the solution set is $\{0, 1\}$. Hence, the equation has two solutions.